

Active Directory Home Lab — DC01 Build

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Version: 1.0

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Introduction

This project documents the step-by-step build of an Active Directory environment as part of my SOC analyst home lab. Active Directory is one of the most important systems in any organization. It controls who can log in, what they can access, and how security rules are applied across all computers in a network.

As a cybersecurity student working toward a SOC Analyst role, understanding Active Directory is essential. Most real-world attacks target Active Directory, so learning how to build, manage, and secure it gives me the hands-on knowledge needed to detect and defend against those attacks.

In this lab, I build a Windows Server 2022 Domain Controller named **DC01** using VMware Workstation on my HP laptop. The domain is named **ezekiel.lab**. Once the Domain Controller is running, I will join Windows 11 client machines to the domain to simulate a real company network.

Tools used:

- VMware Workstation (virtualization)
- Windows Server 2022 (Domain Controller)
- Windows 11 Enterprise (client machines)

What this lab demonstrates:

- Installing and configuring a Domain Controller
- Creating and managing a domain
- Joining client computers to a domain
- Building a foundation for SOC monitoring and detection

STEP 1 — Create the Virtual Machine

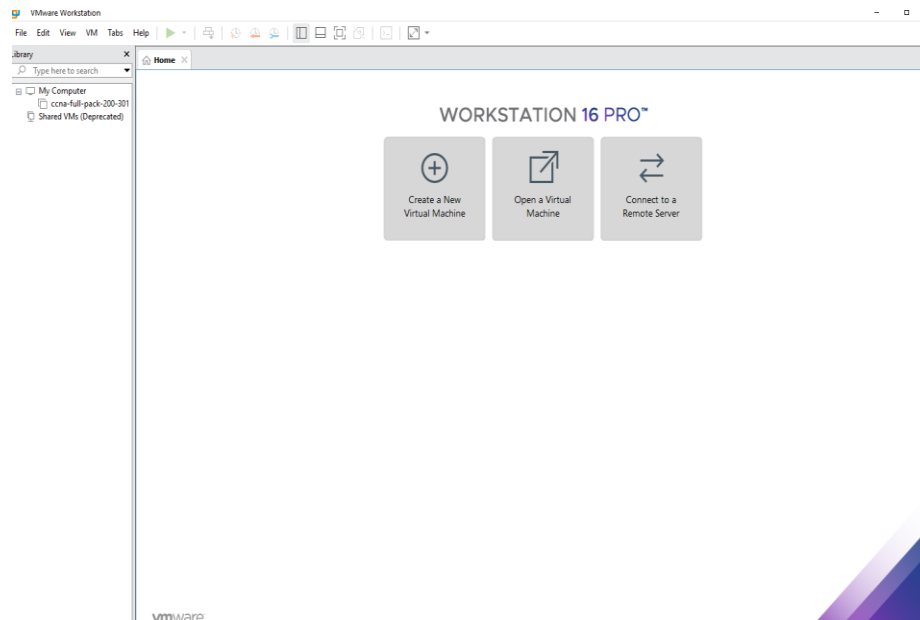


Figure 1 — VMware Workstation home screen



Figure 2 — New Virtual Machine Wizard, selecting Typical configuration

click **"Next"**

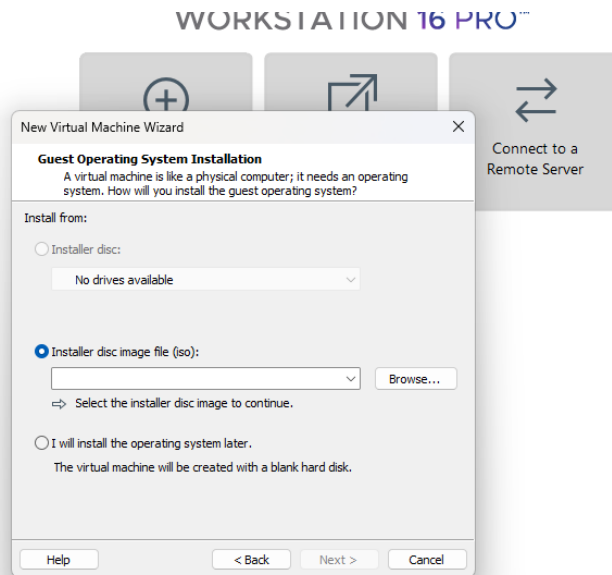


Figure 3 — Choosing the installer disc image (ISO) file

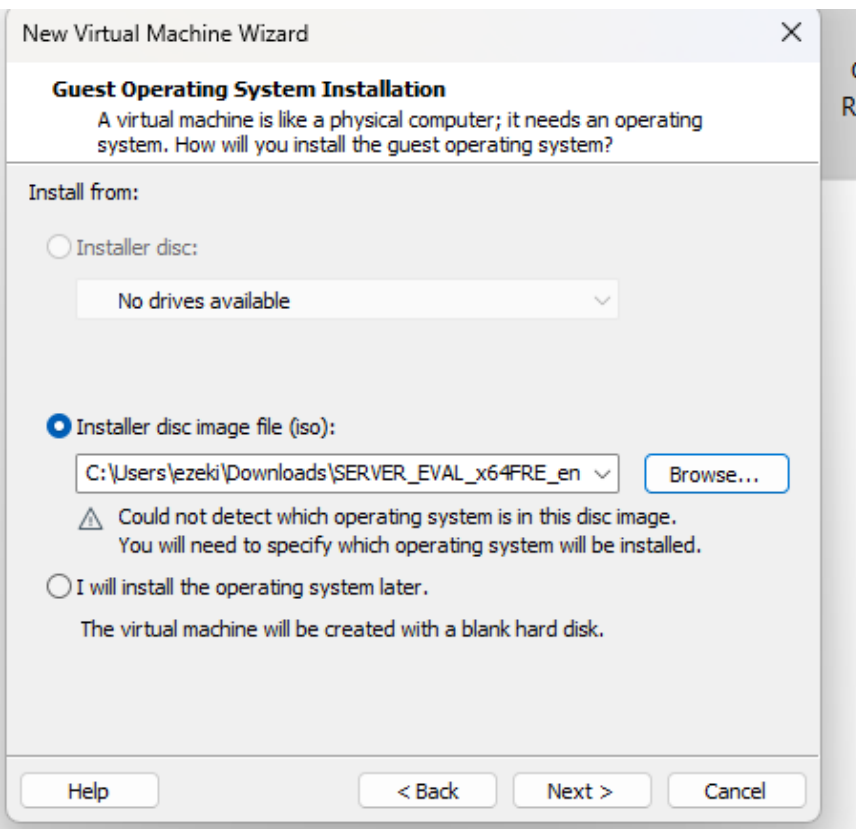
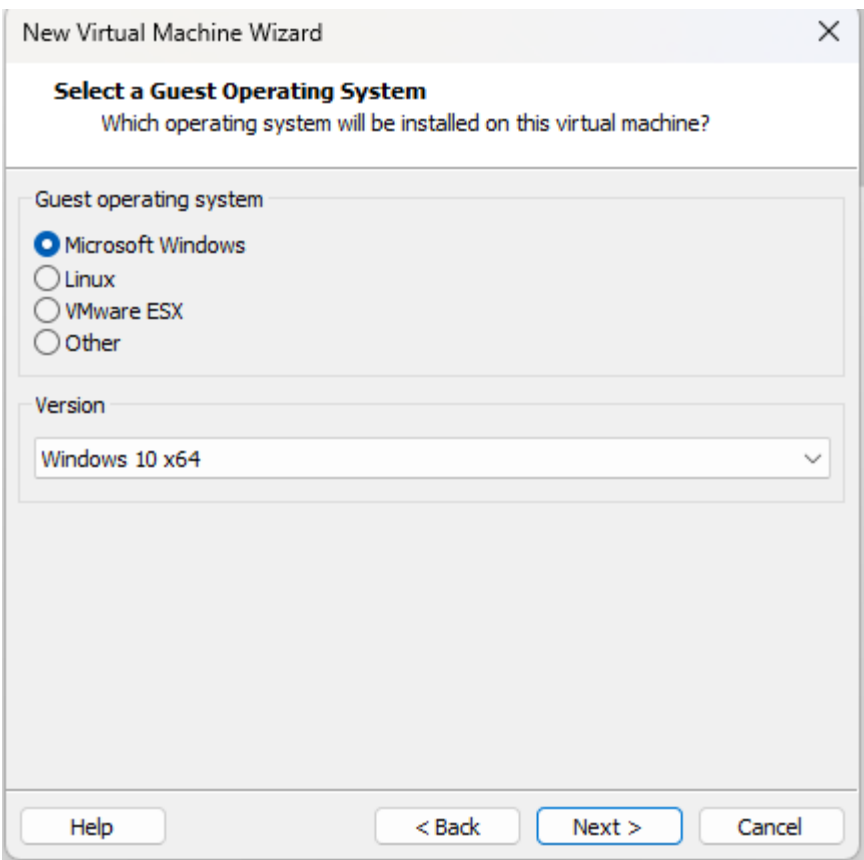
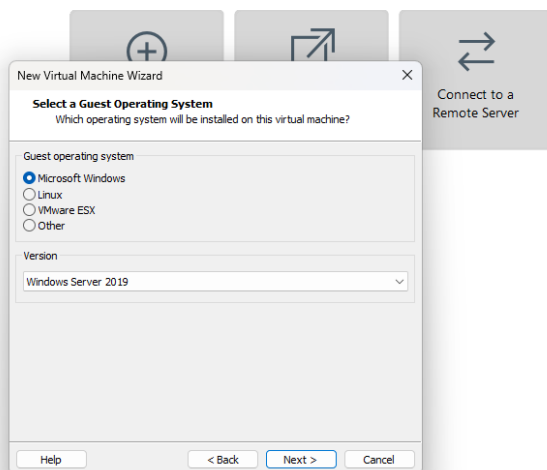


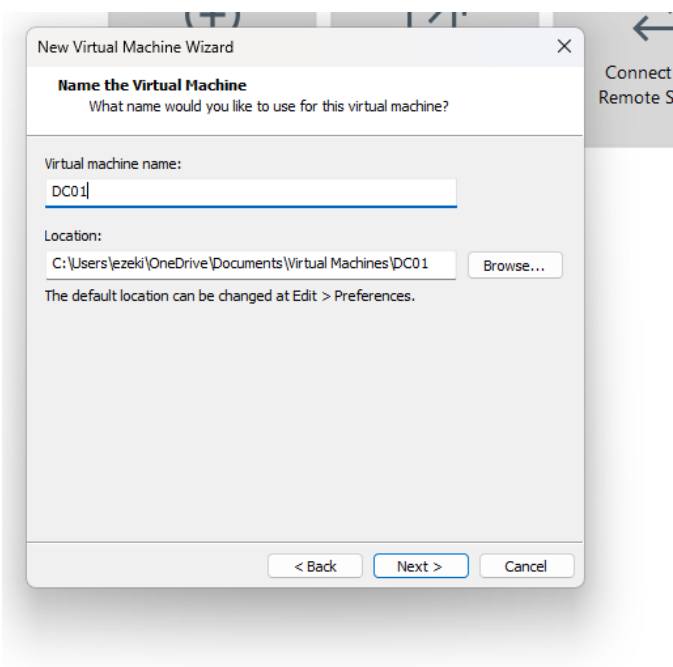
Figure 4 — Windows Server 2022 ISO loaded into VMware



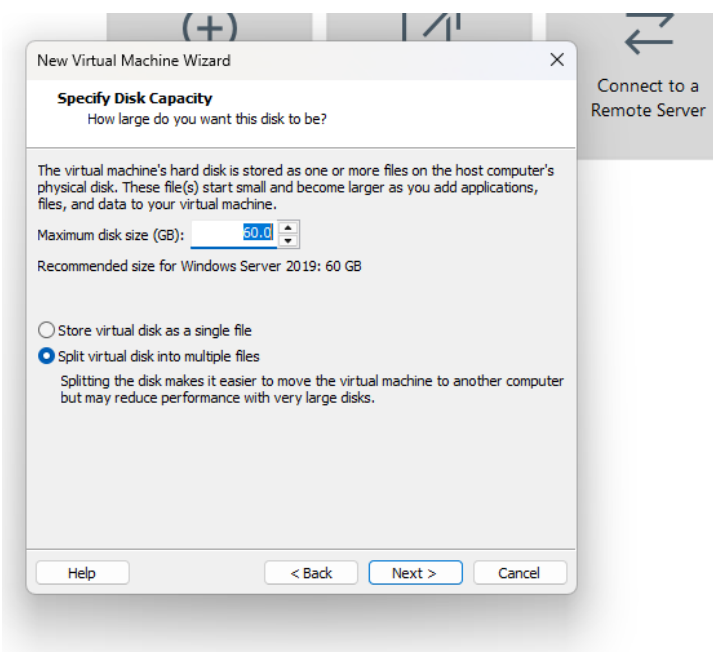
Note: I selected Microsoft Windows as the guest operating system and chose Windows Server 2019 as the version. Even though I am installing Windows Server 2022, the 2022 option was not available in this version of VMware. Selecting 2019 works fine because VMware only uses this setting to apply default hardware settings — it does not affect the actual installation.



Naming the Virtual Machine

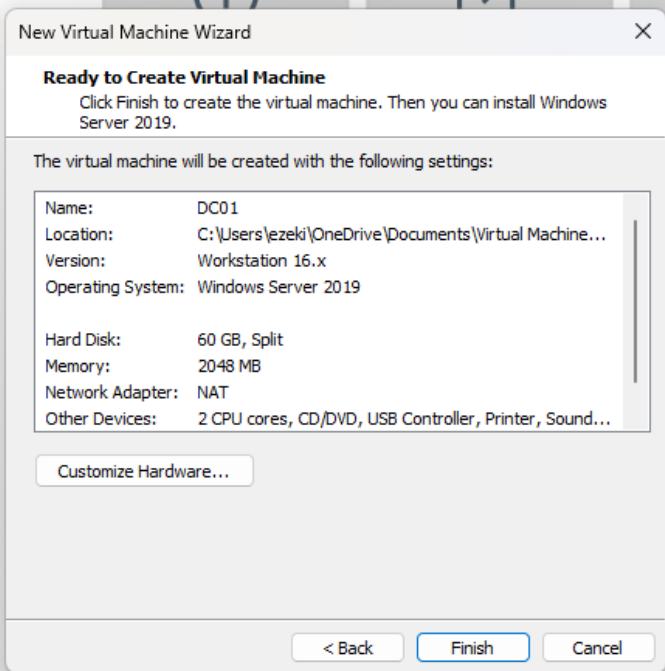


Note: I named the virtual machine DC01, which stands for Domain Controller 01. This is the main server that will run Active Directory and control the entire domain.



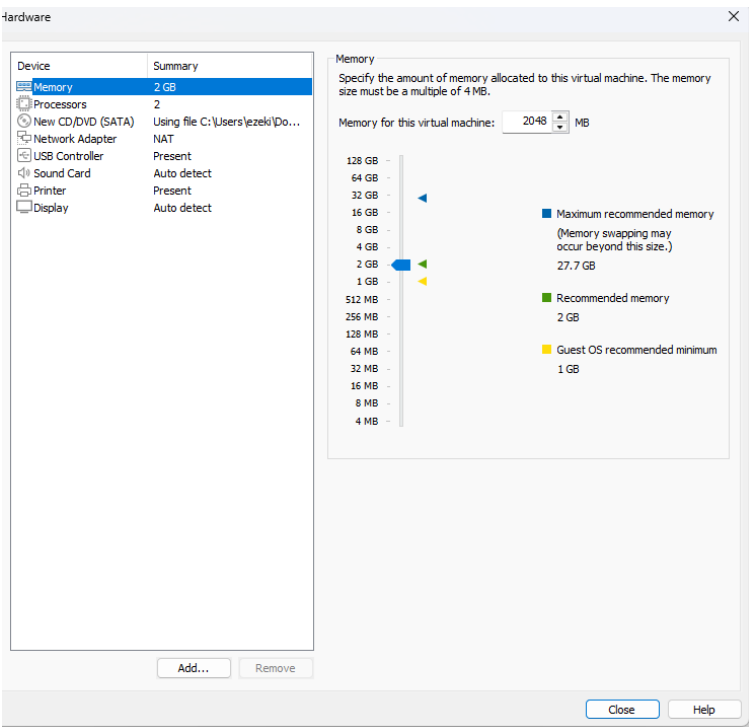
Setting the Disk Size

Note: I set the virtual hard disk to 60 GB, which is the recommended size for Windows Server. This gives the Domain Controller enough space to run Active Directory and store user data.

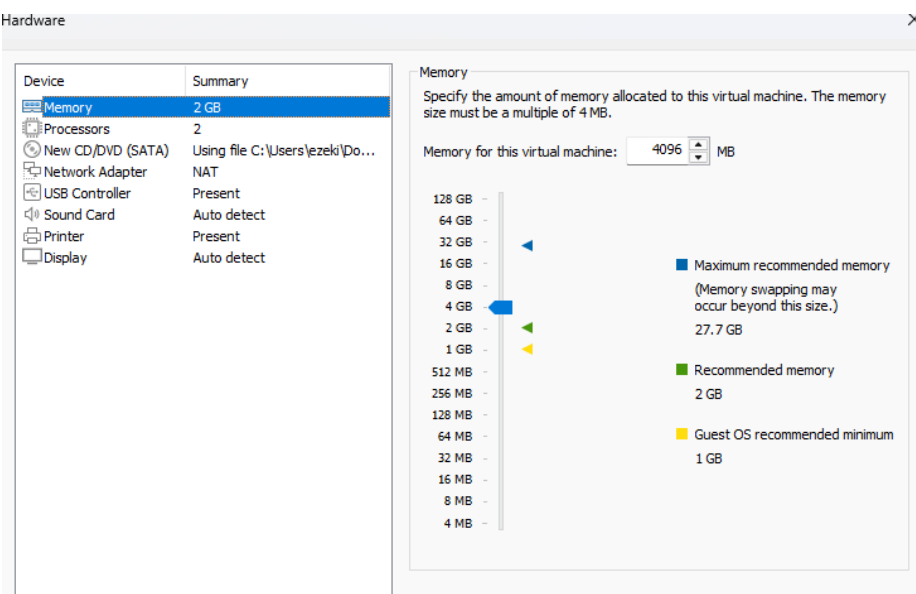


Virtual machine summary before creation

Note VMware set the default memory to 2GB. I increased it to 4096 MB (4GB) to match the lab design requirements

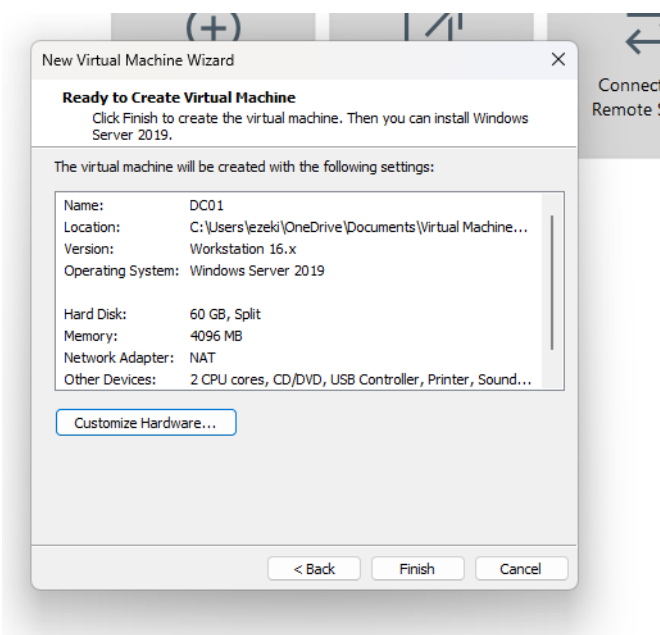


Adjusting the Memory (RAM)

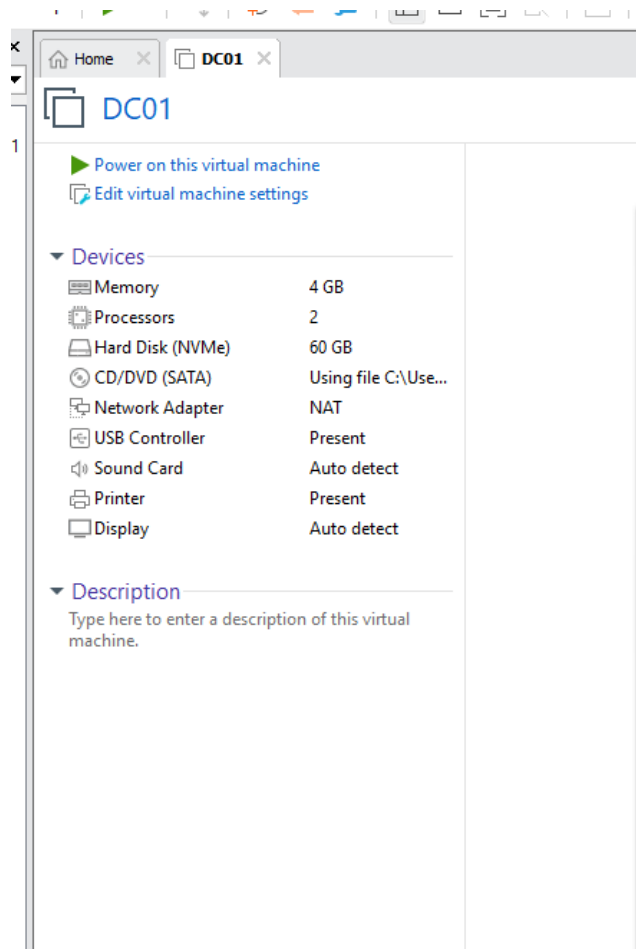


Setting the virtual machine memory to 4GB

Note: VMware set the default memory to 2GB, but my lab documentation specifies 4GB for the Domain Controller. I increased the memory to 4096 MB (4GB) so the build matches my planned design. Active Directory runs DNS and DHCP services, so it needs enough memory to perform well.



Virtual Machine Created Successfully



DC01 virtual machine created and ready to power on

Note: The DC01 virtual machine has been successfully created with 4GB RAM, 60GB storage, and 2 processors. The machine is now ready to power on and begin the Windows Server installation

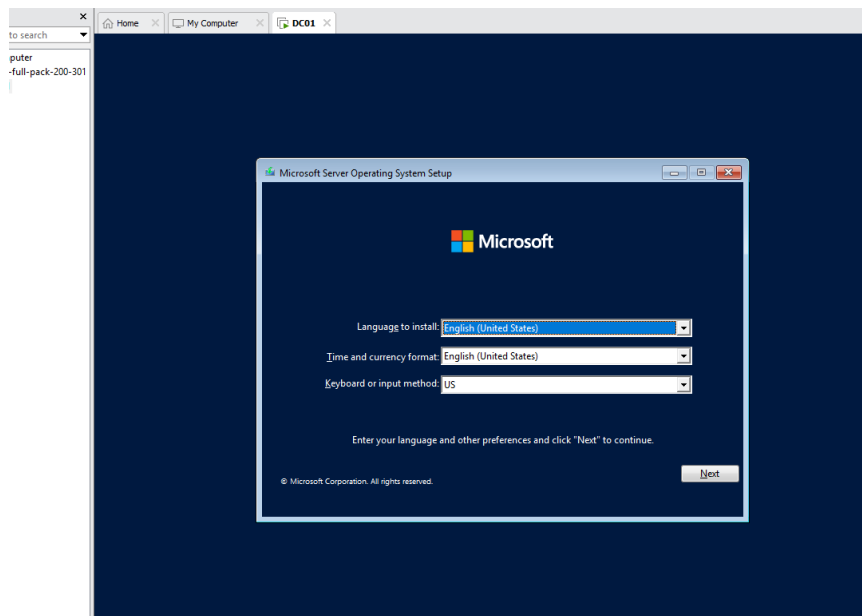
Challenge: VMware 16 crashed with access violation errors. Solution: Diagnosed a Windows 11 / Hyper-V compatibility conflict and upgraded to VMware Workstation 26H1, which resolved it.

Challenge: VMware Workstation 16 repeatedly crashed with "unrecoverable error: Exception 0xc0000005 (access violation)" when powering on the VM. The Windows Server logo would appear for a few seconds, then the VM powered off.

Investigation: I verified the ISO integrity using SHA256 (it was genuine), checked BIOS virtualization (already enabled), disabled Memory Integrity, and disabled Secure Boot. The crash continued.

Root Cause: Research revealed VMware Workstation 16 has a known compatibility conflict with Windows 11's Hyper-V and Credential Guard security features.

Solution: I upgraded to VMware Workstation 26H1, which is built to work with Windows 11's security features. The crash was resolved and Windows Server installed successfully.



Windows Server installation begins

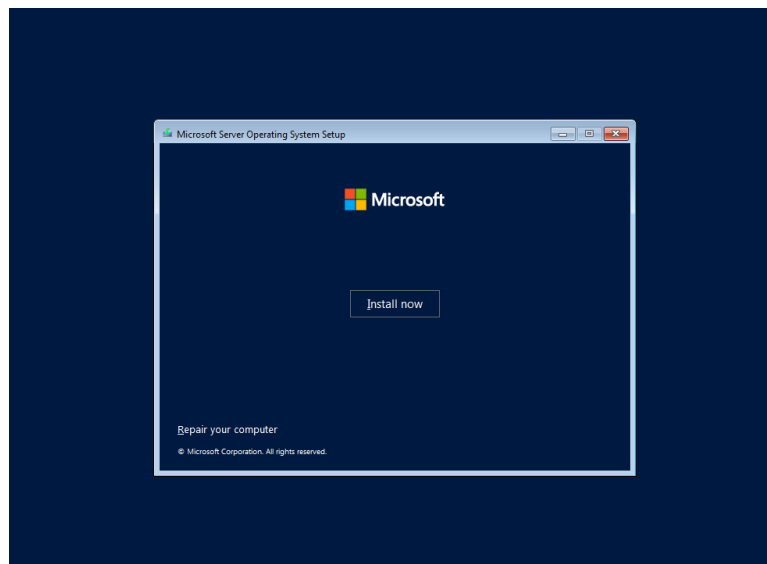


Figure — Beginning Windows Server 2022 installation

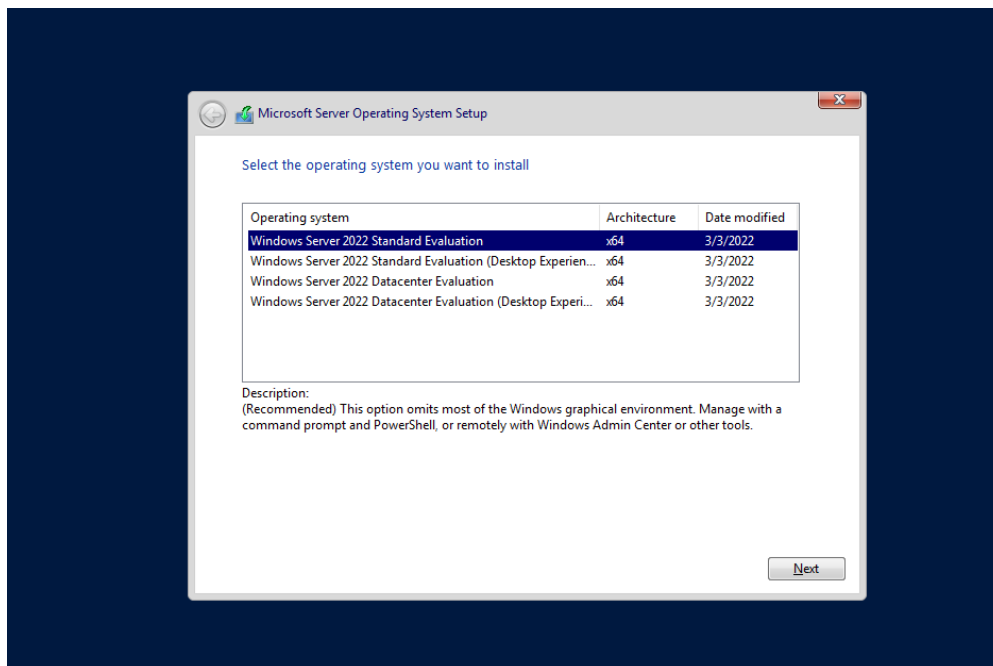


Figure — Choosing Windows Server 2022 Standard Evaluation (Desktop Experience)

Note: I selected "Windows Server 2022 Standard Evaluation (Desktop Experience)." The Desktop Experience option includes the full graphical interface with a mouse and desktop, which is easier to manage than the Server Core option that only uses a command line. Standard edition is sufficient for a home lab Domain Controller, Datacenter edition is designed for large enterprise environments

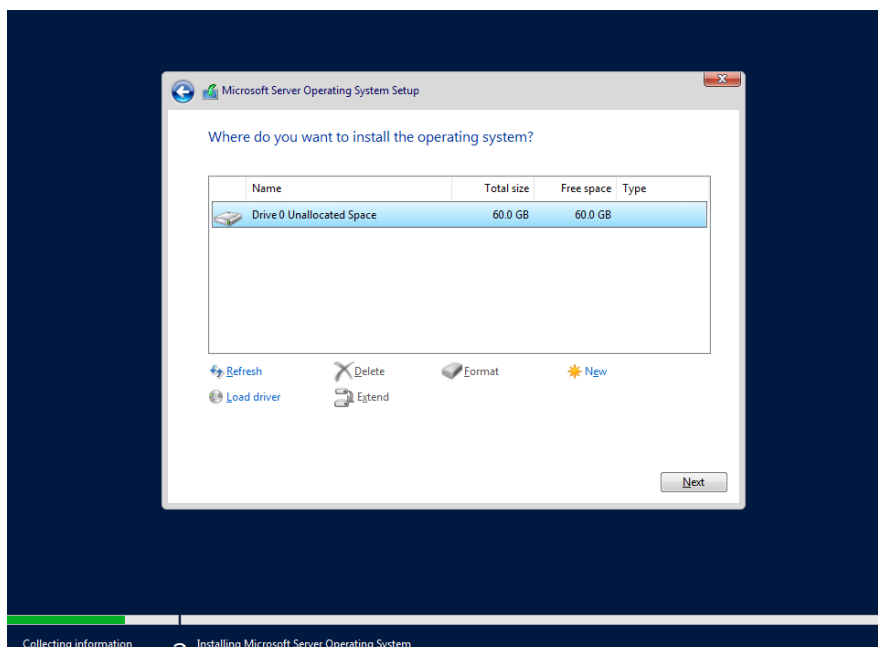


Figure — Selecting Custom (clean install)

Note: I chose "Custom: Install Microsoft Server Operating System only" because this is a fresh installation on an empty disk. The Upgrade option is only used when an operating system is already installed and you want to keep existing files. Since DC01 is a brand new virtual machine, Custom is the correct choice.

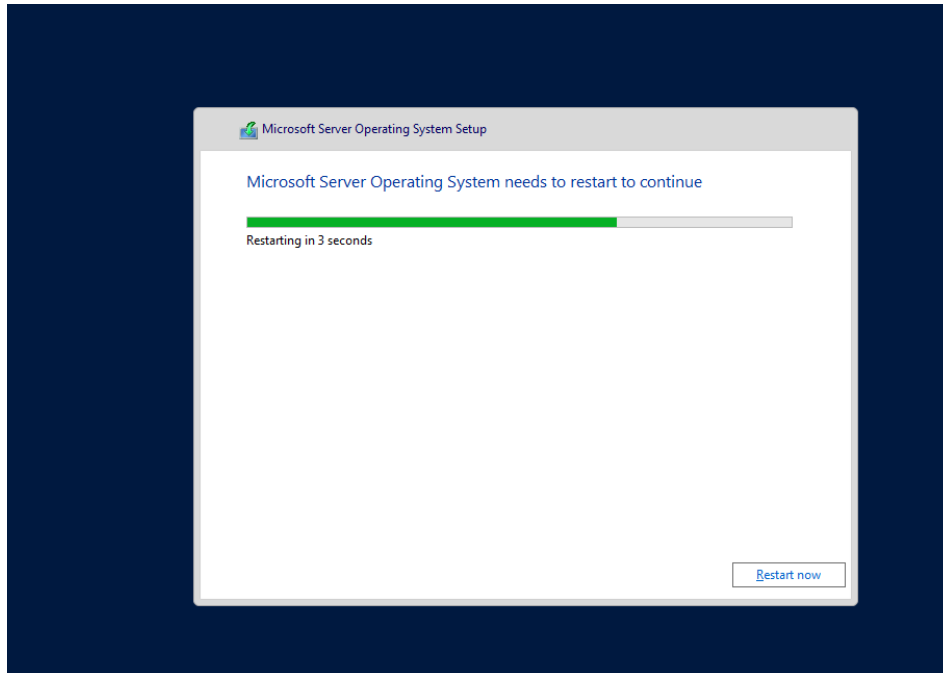
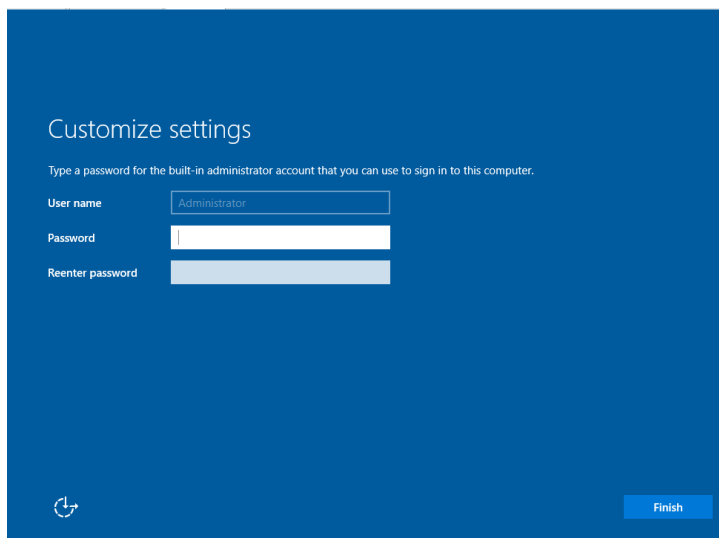
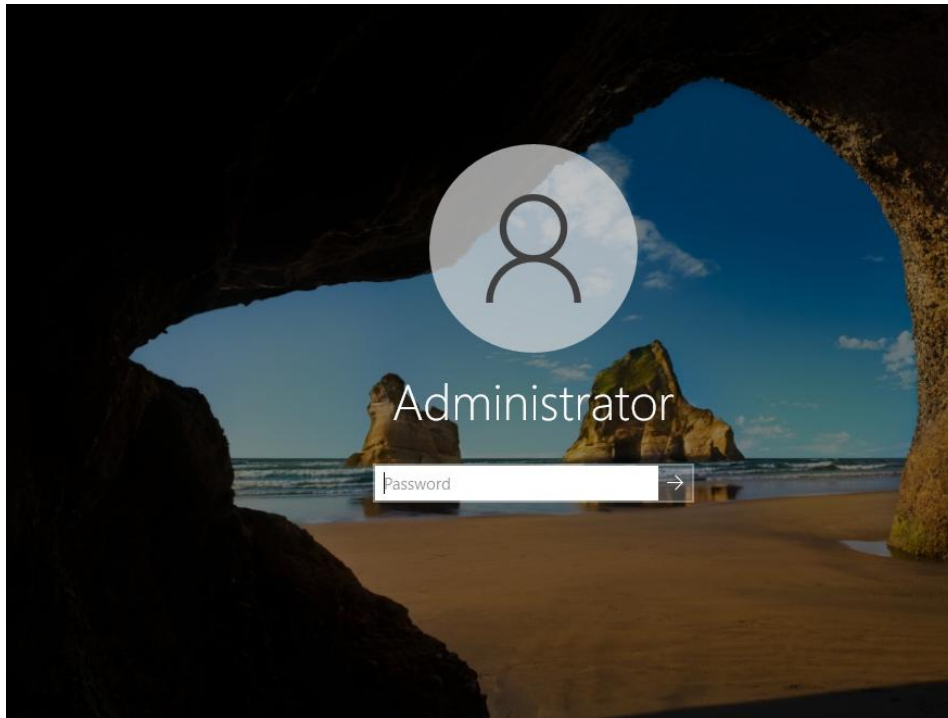


Figure — Windows Server 2022 installing successfully

Note: The installation began successfully, copying files and configuring Windows Server 2022. This confirmed the VMware upgrade fully resolved the earlier crash issues.



Note: I set a strong Administrator password meeting Windows complexity requirements (8+ characters, uppercase, number, symbol). For security, the actual password is stored privately, never in documentation.



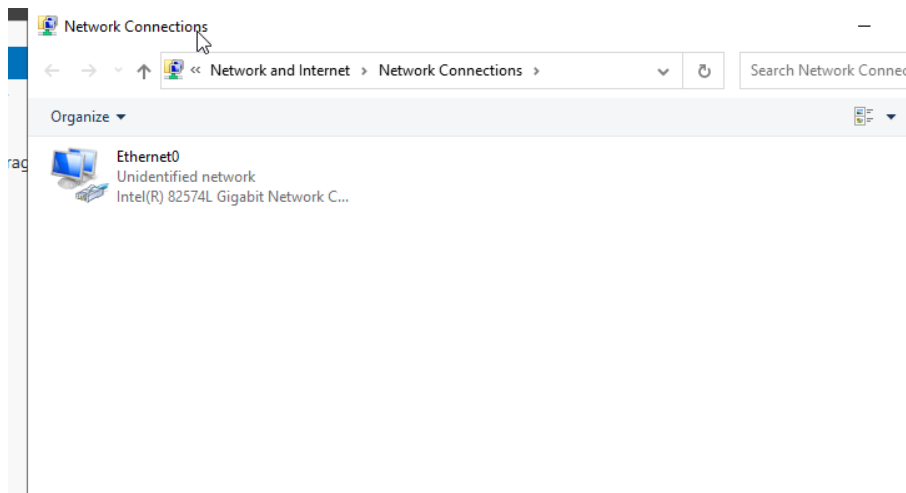
Configuring Static IP Address on DC01

Why we do this:

A Domain Controller must always have the same IP address. If the IP changes, Active Directory, DNS and all client machines lose connection.

IP settings from lab documentation:

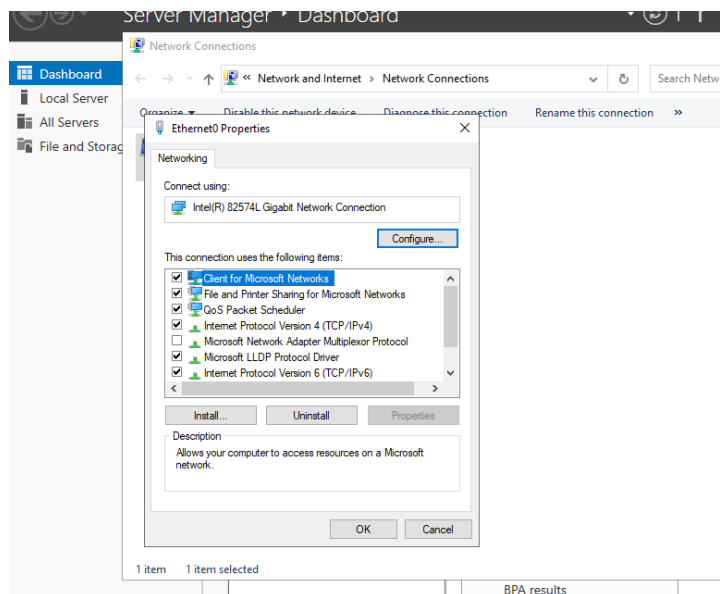
- IP Address: **192.168.10.10**
- Subnet Mask: **255.255.255.0**
- Default Gateway: **192.168.10.1**
- DNS: **127.0.0.1**



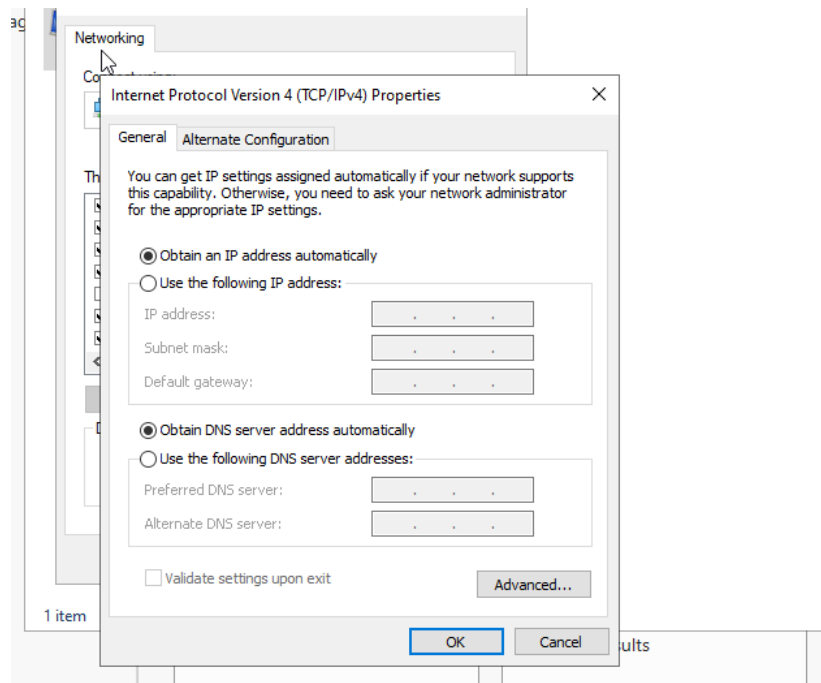
Configure Static IP Address

Note:

To give DC01 a fixed IP address, I opened Network Connections by running `ncpa.cpl`. I right-clicked Ethernet0 and opened Properties. Ethernet0 is the virtual network adapter created by VMware for DC01

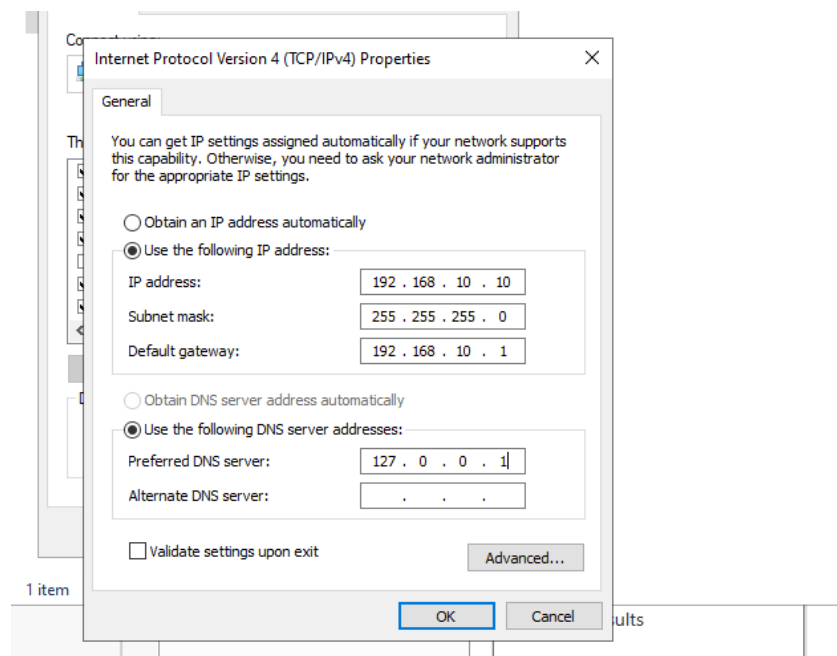


TCP IPv4 Properties



Note:

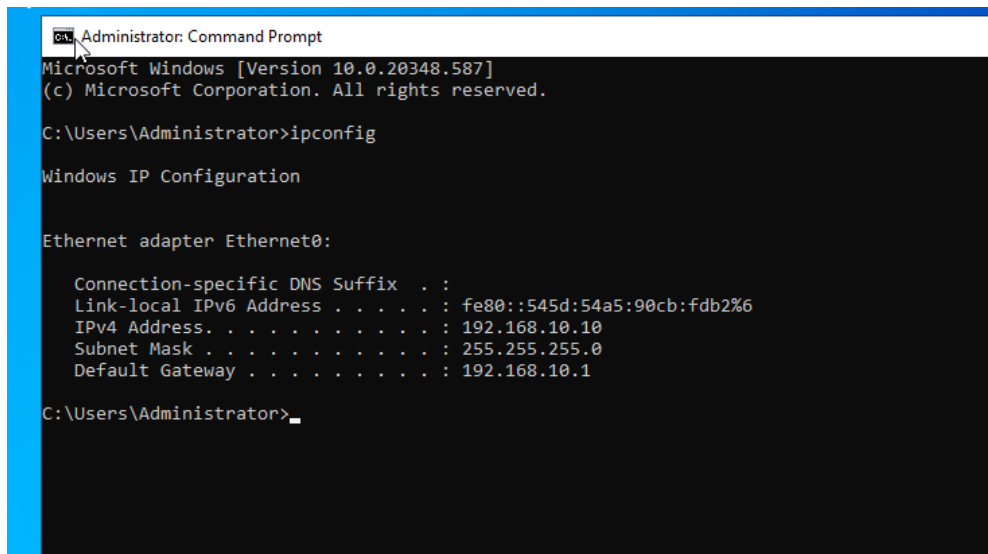
By default, DC01 was set to obtain an IP address automatically (DHCP). A Domain Controller must have a static IP address that never changes. This is because other computers on the network need to always find DC01 at the same address to log in and access the domain.



Note:

I configured DC01 with a static IP address. A Domain Controller must have a fixed IP address so that all computers on the network can always find it at the same location. The settings are:

- IP Address: 192.168.10.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1
- Preferred DNS: 127.0.0.1 (points to itself because DC01 will host DNS)



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

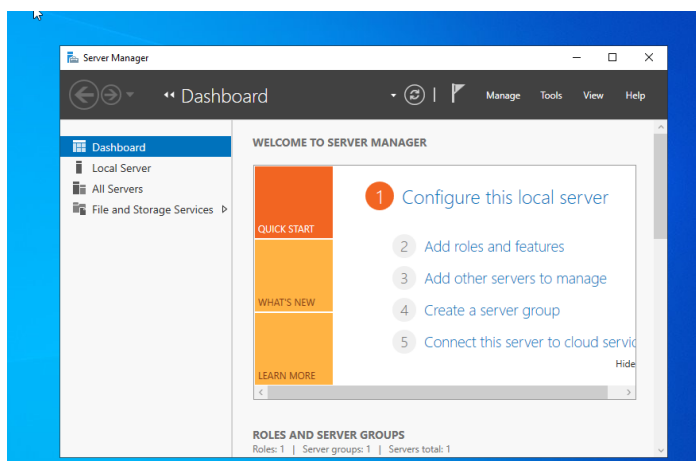
    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::545d:54a5:90cb:fdb2%6
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.10.1

C:\Users\Administrator>
```

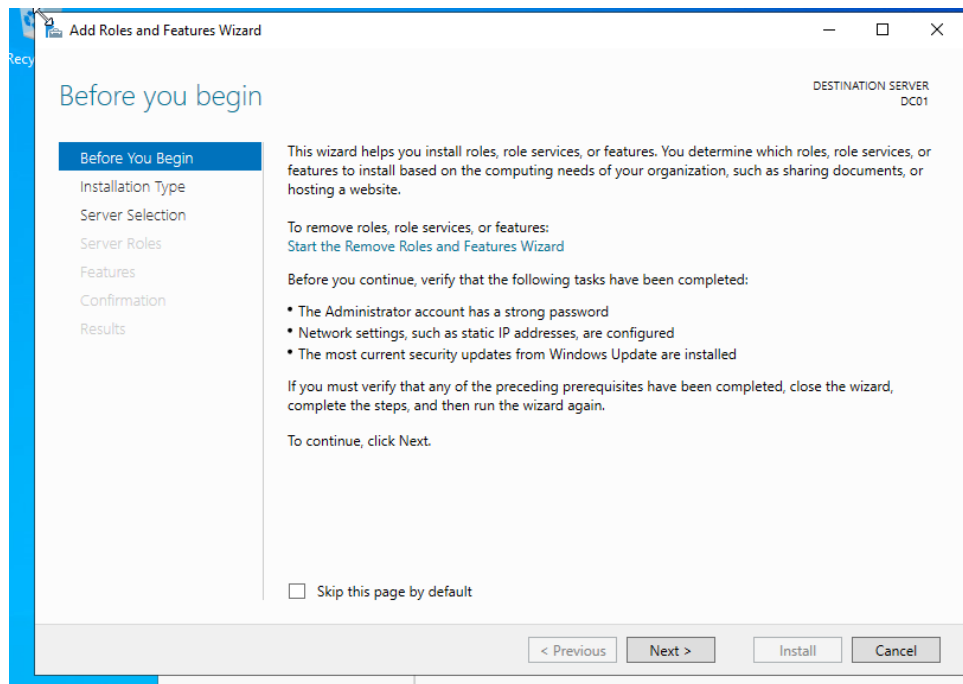
Note:

I ran the command ipconfig in Command Prompt to verify the static IP address was applied correctly. The output confirmed DC01 is now using IP address 192.168.10.10 with subnet mask 255.255.255.0 and gateway 192.168.10.1.

Promote DC01 to Domain Controller

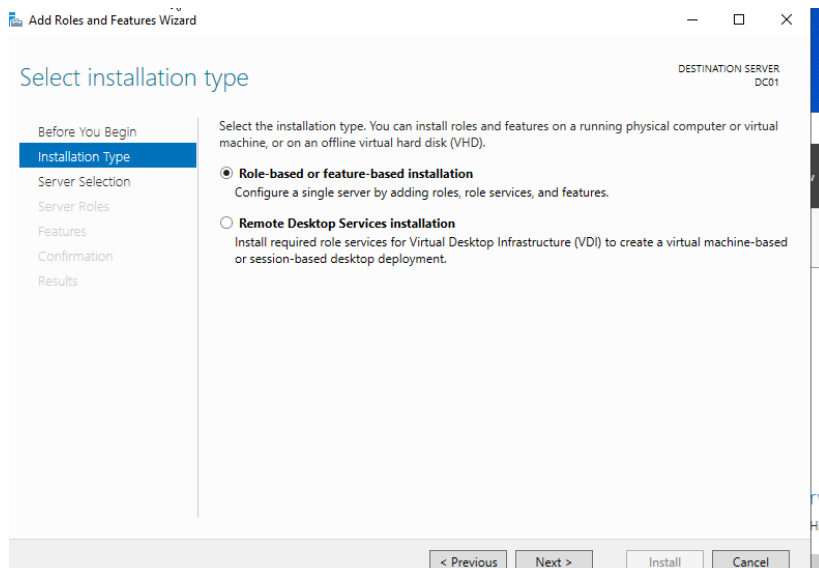


Server Manager Dashboard



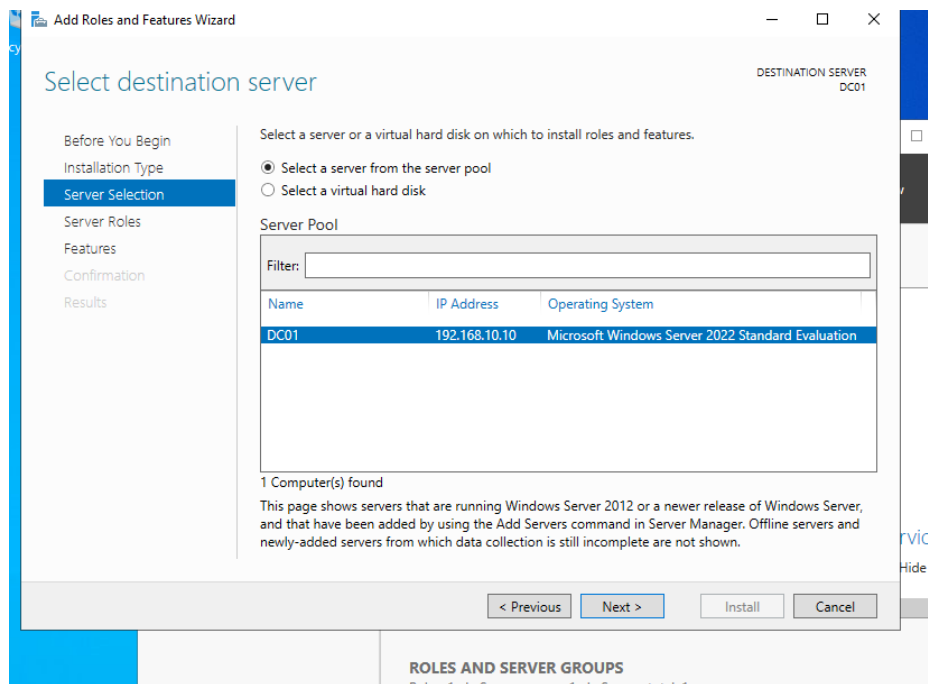
Note:

I opened the Add Roles and Features Wizard from Server Manager. This wizard is used to install the Active Directory Domain Services role on DC01. Before continuing, the wizard confirmed that all prerequisites were already complete strong password, static IP, and server name were all set.

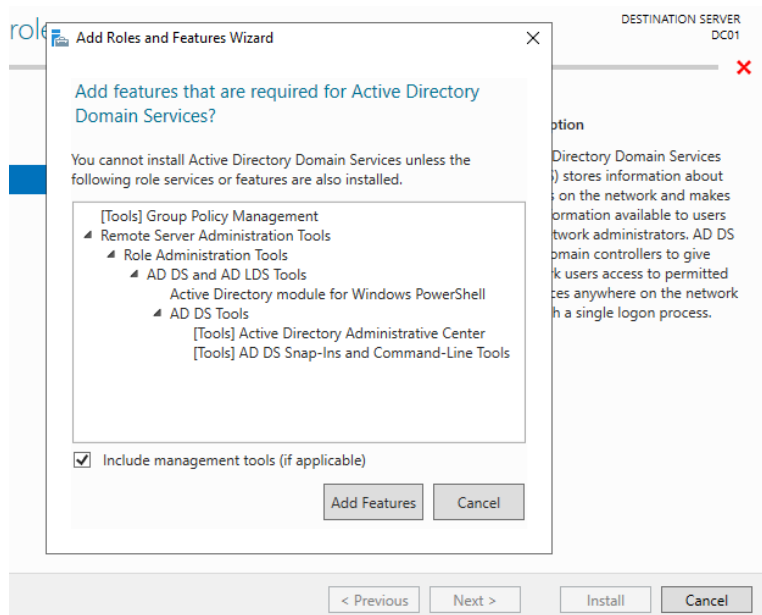


Note: I selected Role-based or feature-based installation. This option installs roles directly on DC01. This is the correct choice for setting up Active Directory on a single server.

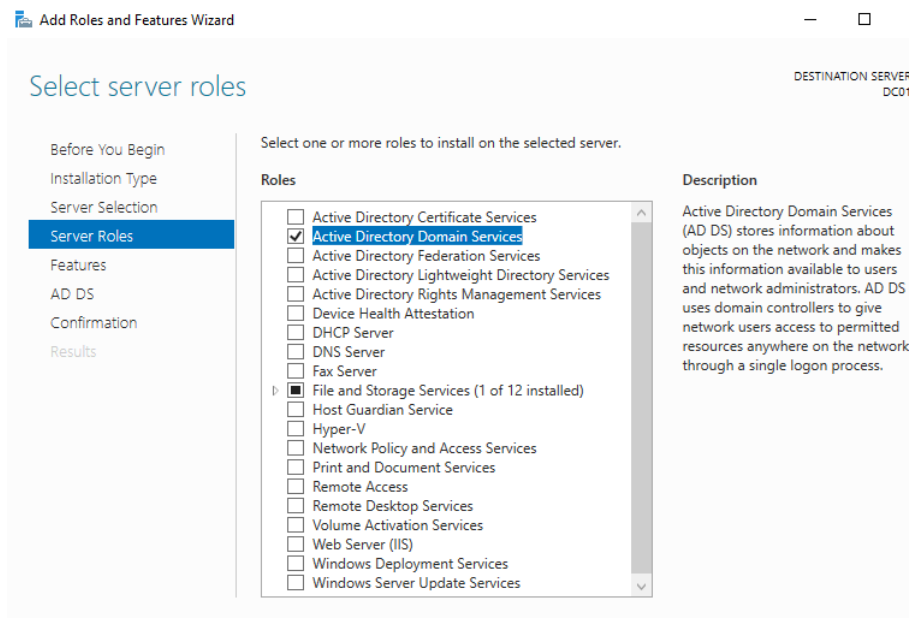
Selecting Role-based or feature-based installation



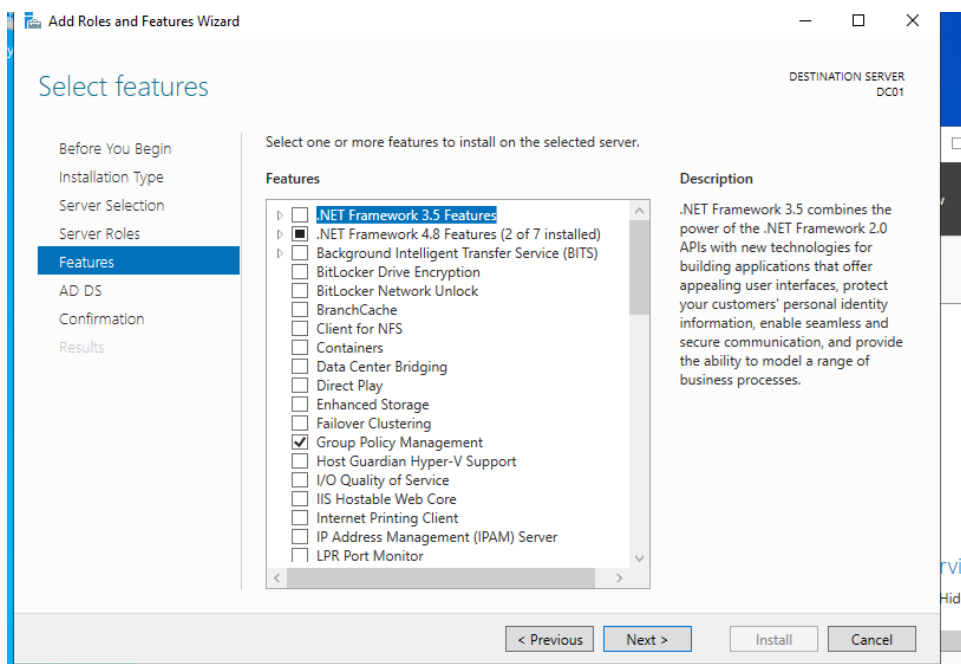
Note: The Server Selection screen confirmed DC01 is visible in the server pool with IP address 192.168.10.10. This proves the static IP address was configured correctly. I selected DC01 as the destination server for the installation.



Note: I selected the Active Directory Domain Services role. A popup appeared asking to add required features including Group Policy Management and AD DS Tools. I clicked Add Features to include all necessary components for Active Directory to work correctly.

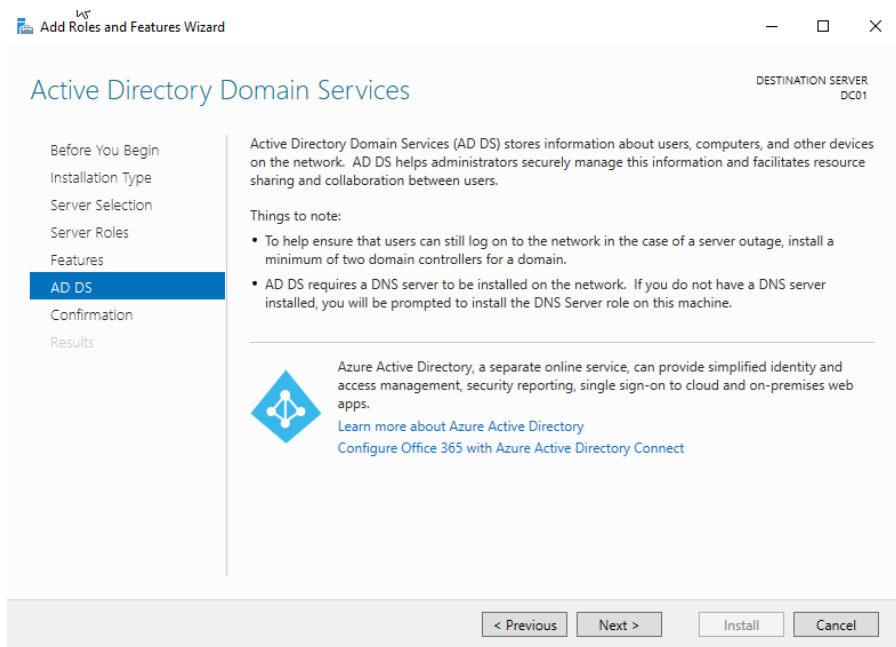


Note: I selected Active Directory Domain Services from the Server Roles list. This is the main role that turns DC01 into a Domain Controller. It allows DC01 to manage users, computers, and security policies across the entire domain.

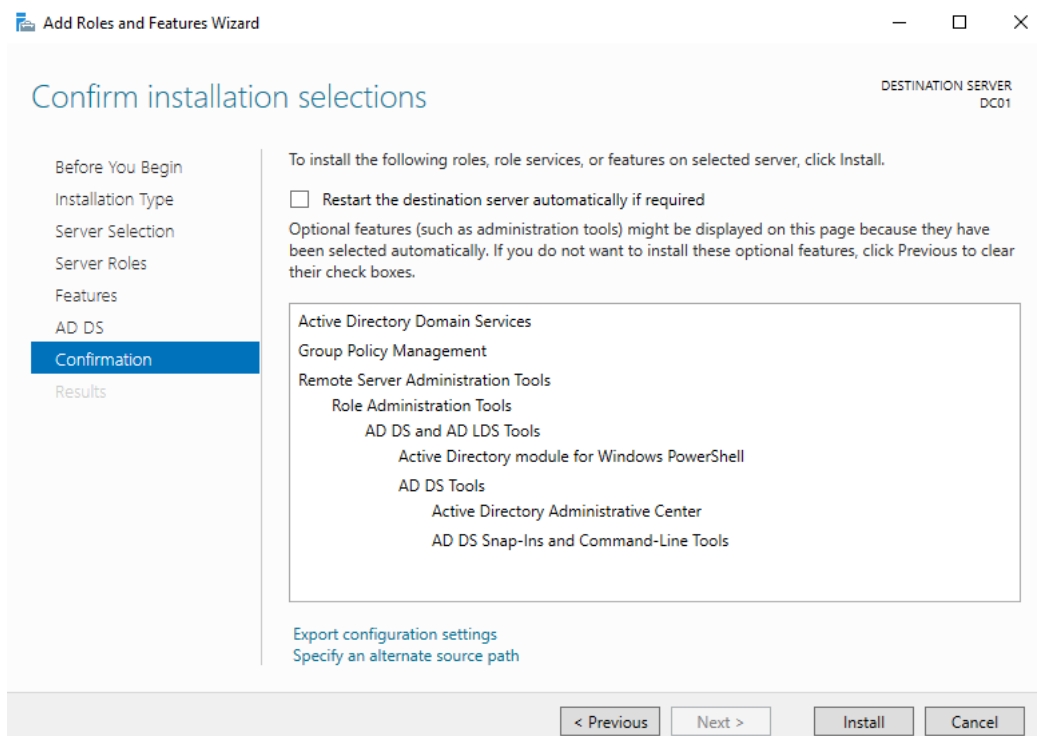


Note:

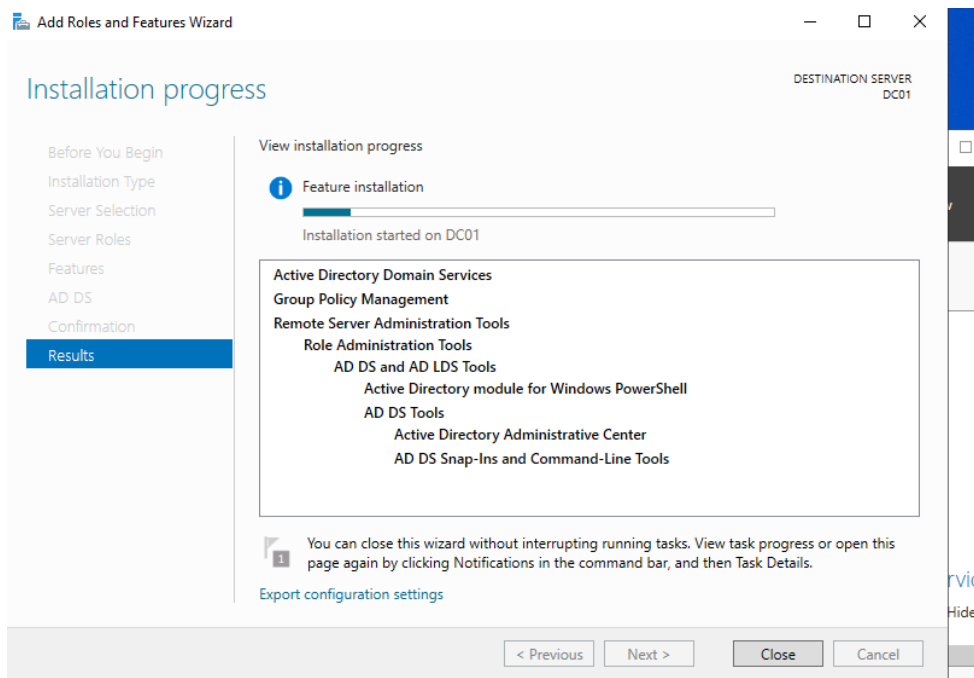
The Features screen showed Group Policy Management already selected. This feature allows administrators to control security settings and policies across all computers in the domain. No additional features were needed at this stage.



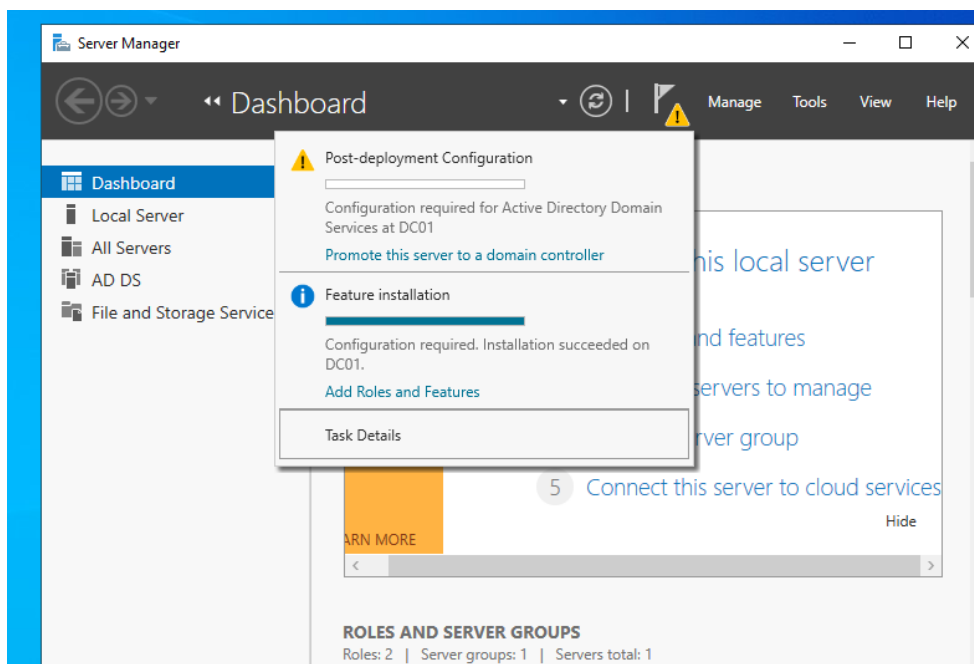
Note: This screen explained what Active Directory Domain Services does. It stores information about all users, computers and devices on the network. It also warned that AD DS requires a DNS server — DNS will be automatically installed when we promote DC01 to a Domain Controller in the next step.



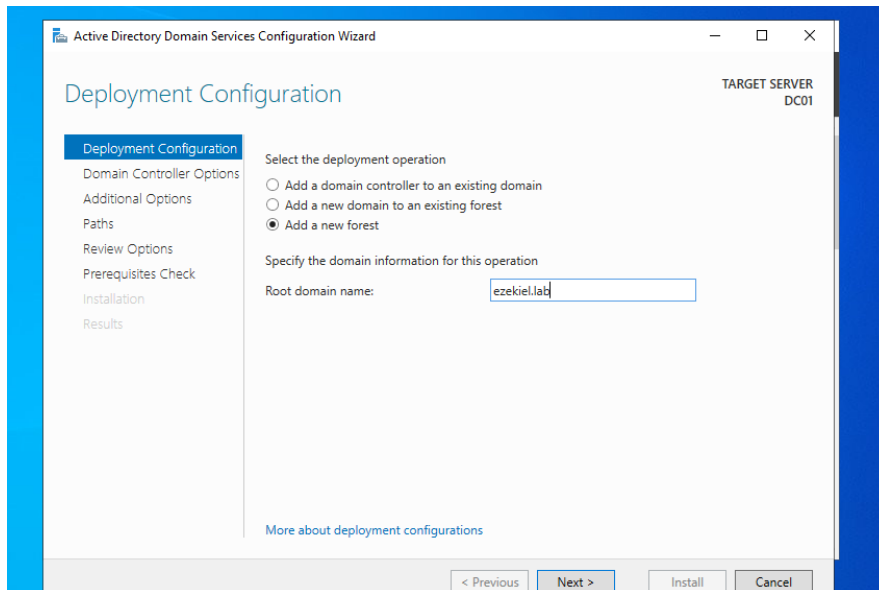
Note: The Confirmation screen showed a summary of everything that will be installed on DC01. This includes Active Directory Domain Services, Group Policy Management, and all required administration tools. I reviewed the list and confirmed it was correct before clicking Install.



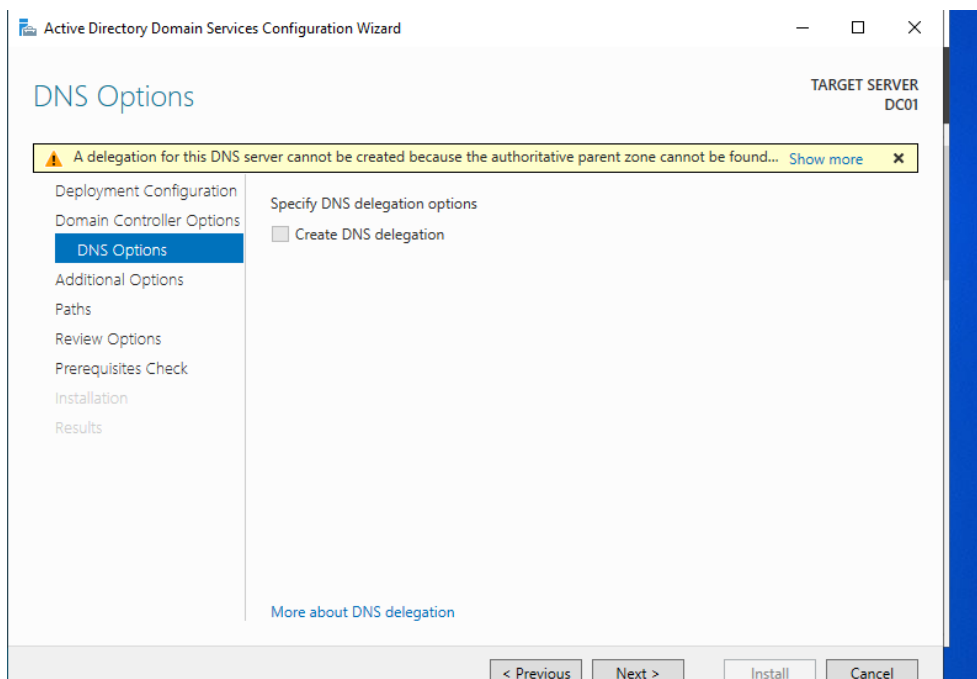
Note: The installation started successfully on DC01. The progress bar confirmed that Active Directory Domain Services and all required components were being installed. This process takes a few minutes to complete.



Note: After the AD DS installation completed, a yellow flag appeared in Server Manager showing a Post-deployment Configuration warning. This is normal. It means AD DS is installed but DC01 still needs to be promoted to a Domain Controller. I clicked "Promote this server to a domain controller" to begin the promotion

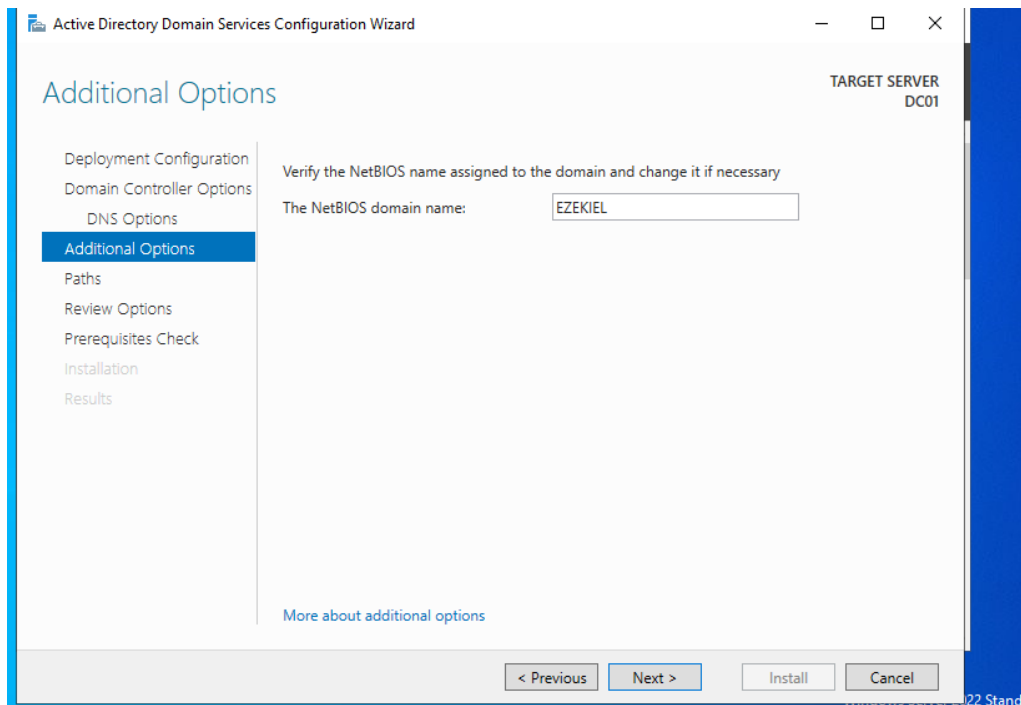


Note: I selected "Add a new forest" because this is a brand new domain with no existing Active Directory. I entered ezeziel.lab as the root domain name. This is the name of the domain that all computers in the lab will join.

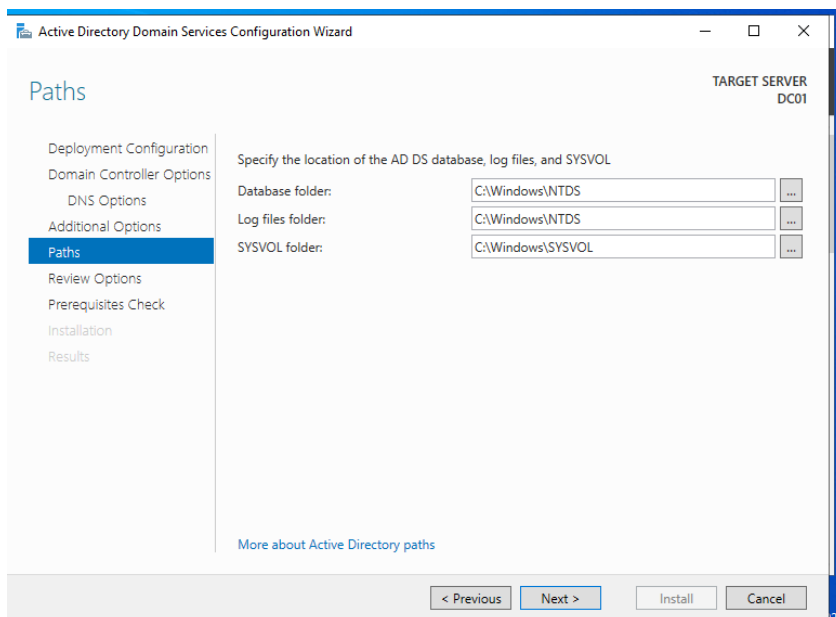


Note:

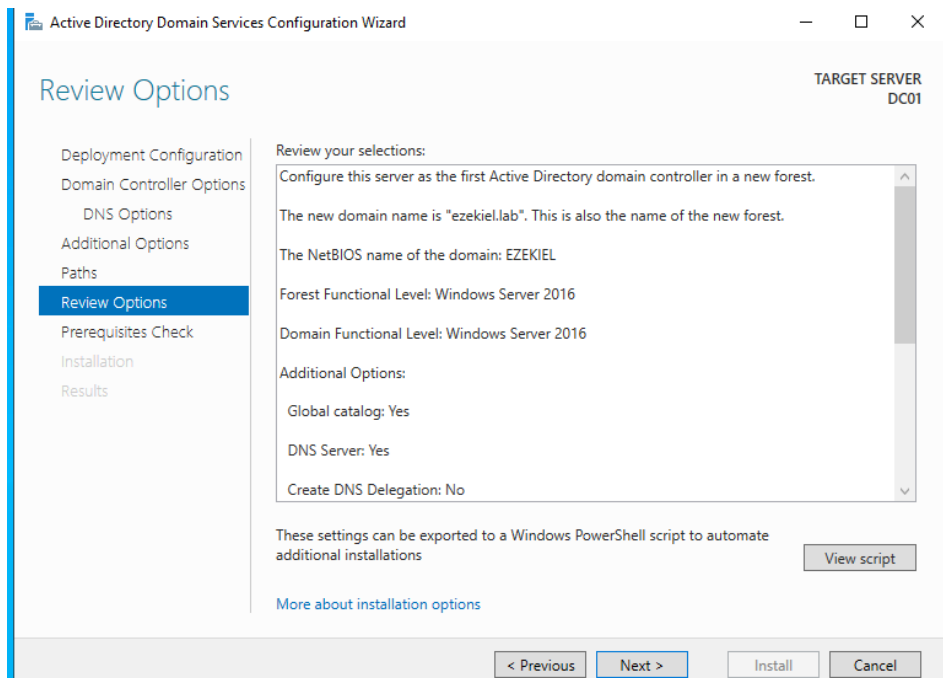
The DNS Options screen showed a warning that a DNS delegation could not be created. This is normal for a new lab domain because there is no parent DNS zone above ezeziel.lab. No action was needed on this screen



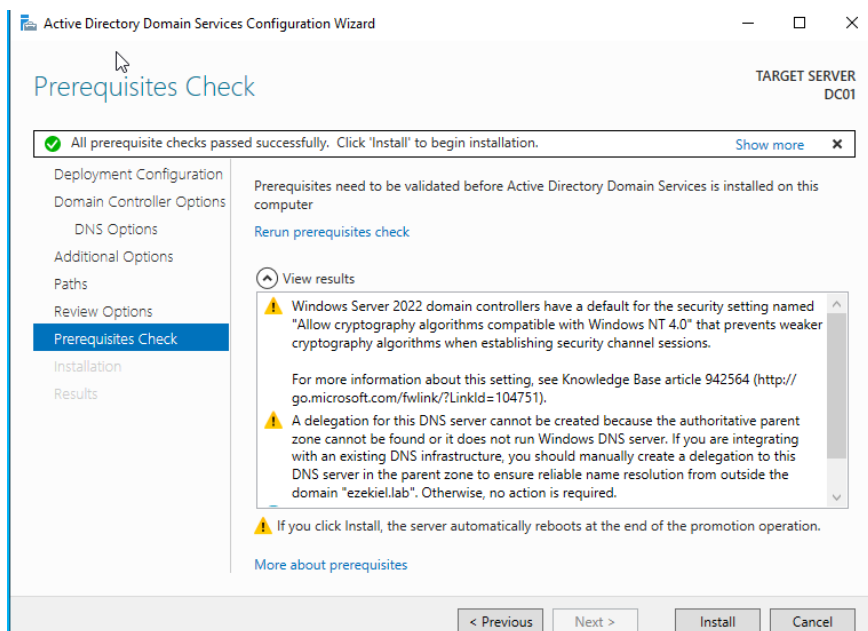
Note:The Additional Options screen showed the NetBIOS domain name as EZEKIEL. NetBIOS is an older naming system used by Windows for backwards compatibility. It was automatically generated from the domain name ezekiel.lab. No changes were needed



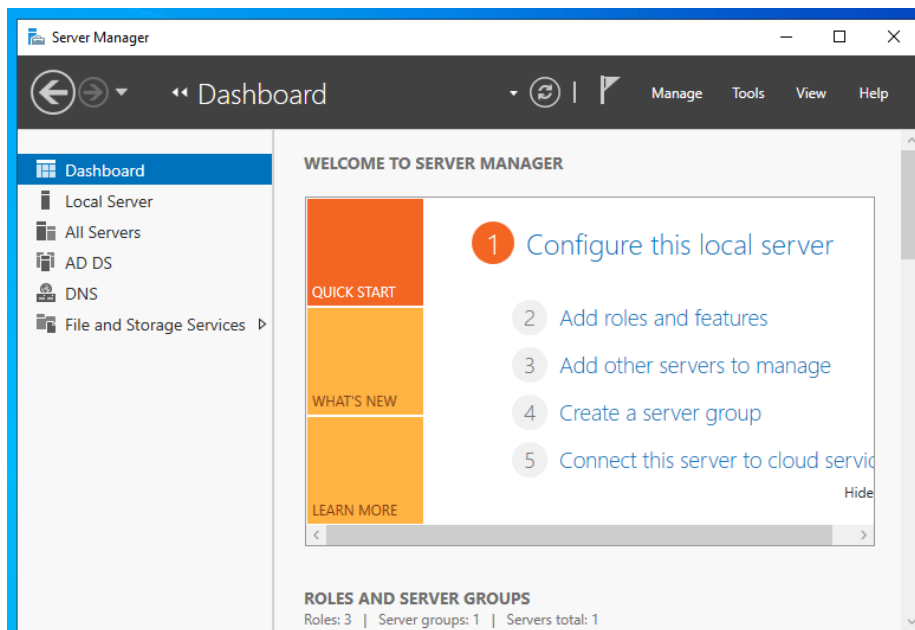
Note:The Paths screen showed where Active Directory will store its database, log files, and SYSVOL folder. All default locations are inside C:\Windows which is the correct location for a lab Domain Controller. No changes were needed.



Note:The Review Options screen confirmed all settings before installation. The domain name is ezeziel.lab, NetBIOS name is EZEKIEL, DNS Server is enabled, and Global Catalog is enabled. I reviewed all settings and confirmed they were correct before proceeding.



Note:The Prerequisites Check passed successfully. The yellow warnings shown are normal for a lab environment. They relate to DNS delegation and cryptography settings which do not affect our lab build. DC01 is ready to be promoted to a Domain Controller.



Note:After DC01 restarted, Server Manager confirmed the promotion was successful. AD DS and DNS now appear in the left menu. This confirms that DC01 is now a fully functioning Domain Controller running the ezekiel.lab domain with DNS installed automatically.

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup ezekiel.lab
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: ::1

Name:     ezekiel.lab
Address:  192.168.10.10

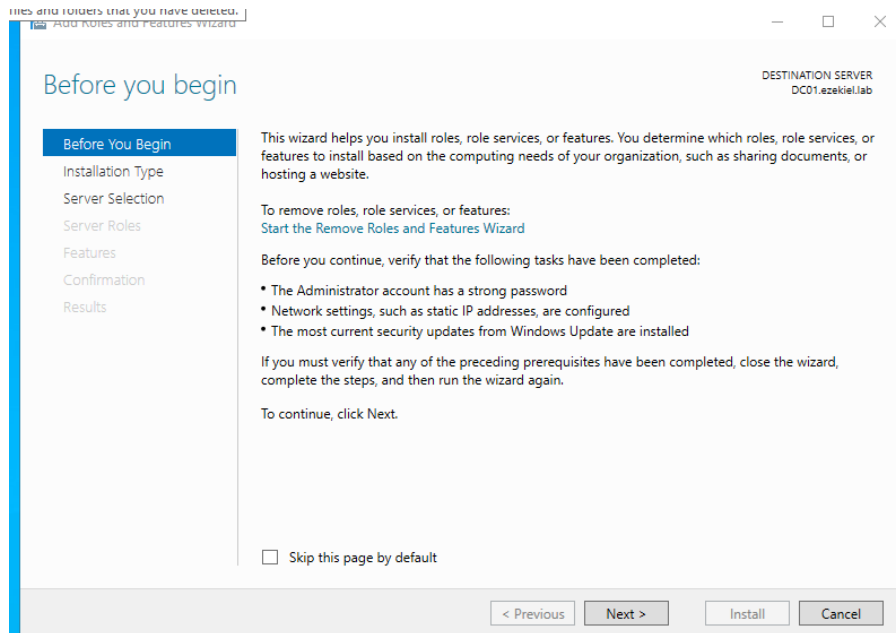
C:\Users\Administrator>
```

Note:I ran the nslookup command to verify that DNS was working correctly. The result confirmed that ezekiel.lab resolves to 192.168.10.10 which is DC01. This proves that the domain and DNS are both working correctly. The timeout warning is normal for a lab environment with no parent DNS zone.

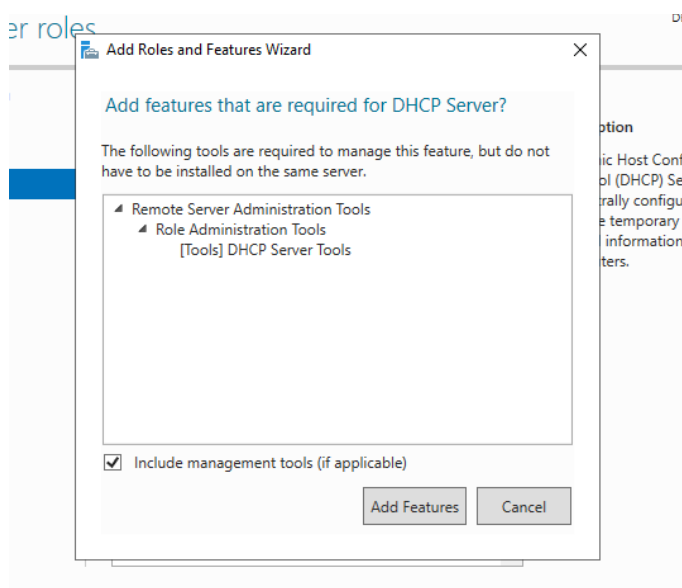
Configure DHCP Server

Note:

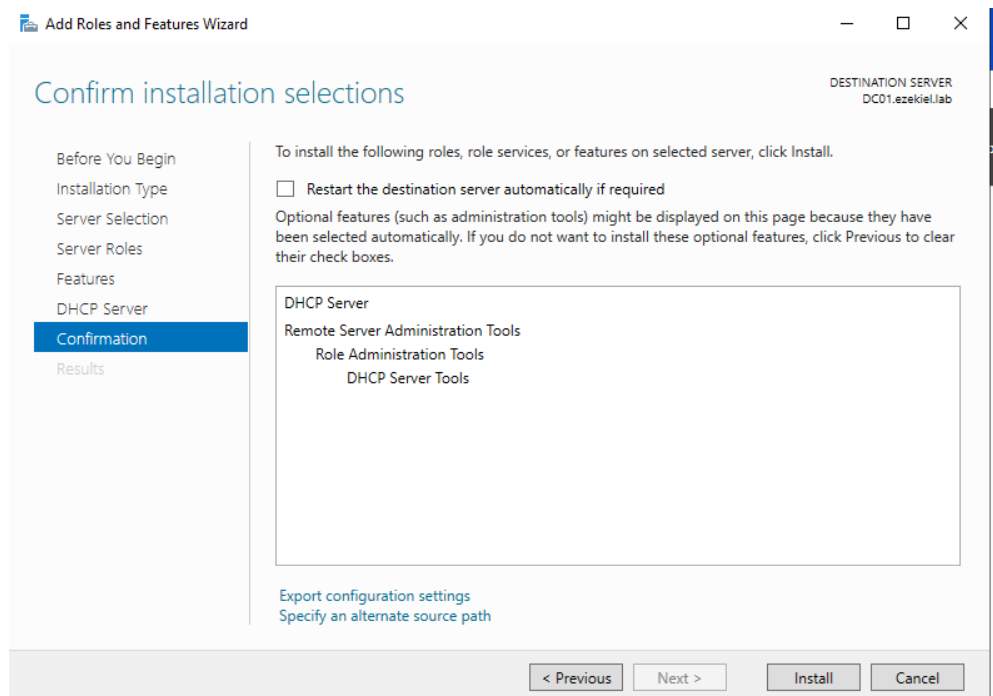
DHCP stands for Dynamic Host Configuration Protocol. It automatically assigns IP addresses to computers that join the ezekiel.lab domain. Without DHCP, every computer would need its IP address set manually. Installing DHCP on DC01 means any new computer that joins the domain will automatically receive an IP address in the 192.168.10.0 range.



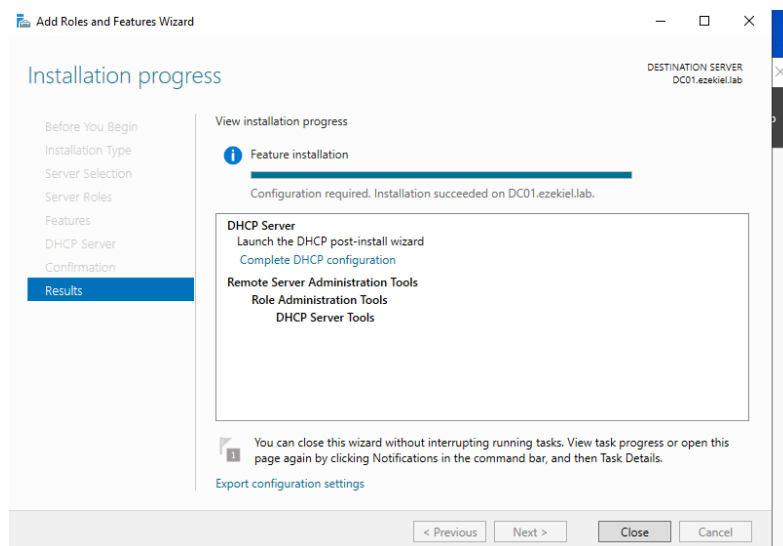
Note: I opened the Add Roles and Features Wizard to install the DHCP Server role. Notice the destination server now shows DC01.ezekiel.lab confirming the domain is active.



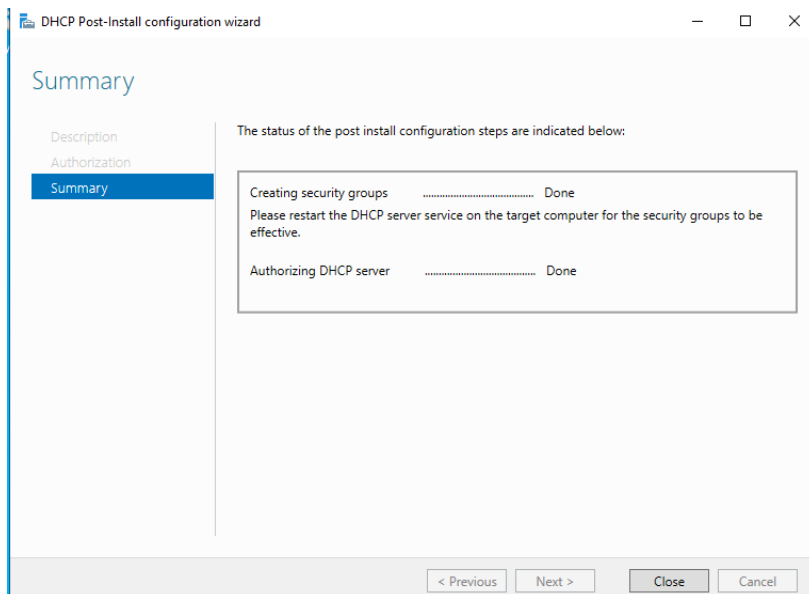
Note: A popup appeared asking to add required features for DHCP Server including DHCP Server Tools. I clicked Add Features to include all necessary management tools for DHCP to work correctly on DC01.



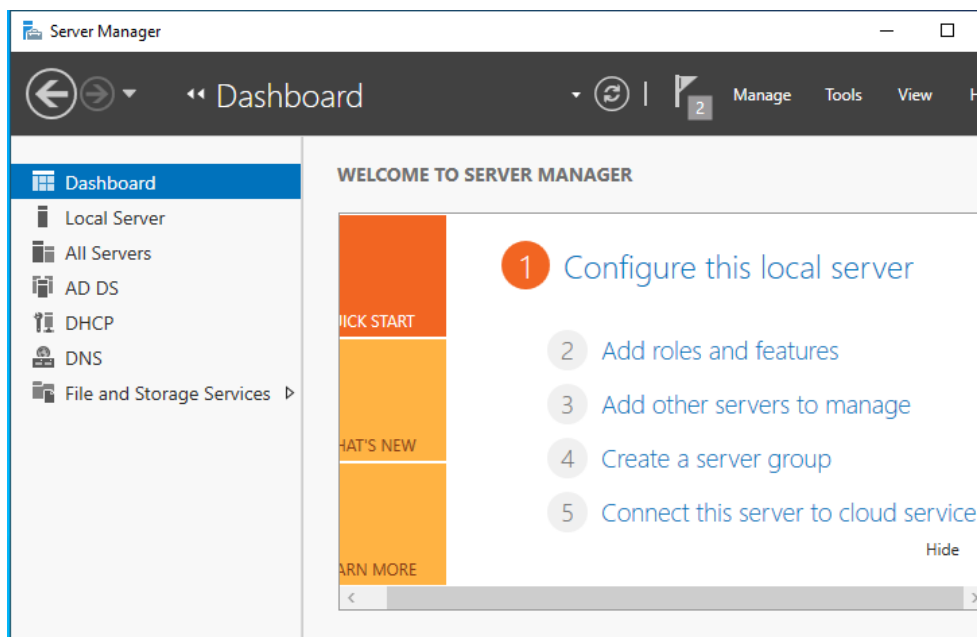
Note: The Confirmation screen showed DHCP Server and DHCP Server Tools ready to install on DC01.ezekiel.lab. I confirmed the selections and clicked Install.



Note: DHCP Server installed successfully on DC01.ezekiel.lab. The installation confirmed that DHCP Server and all management tools were installed without errors.



Note:The DHCP Post-Install wizard completed successfully. Security groups were created and the DHCP server was authorized in Active Directory. This allows DC01 to automatically assign IP addresses to computers that join the ezekiel.lab domain.



DC01 Build Complete

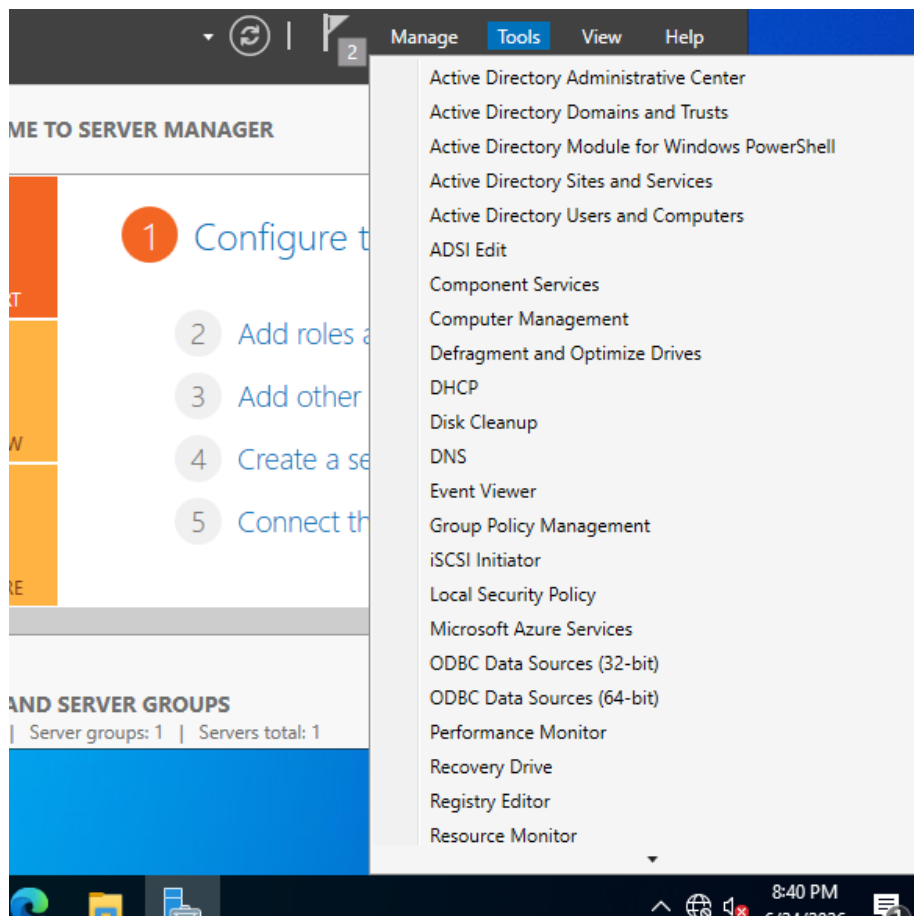
Note:

DC01 is now fully configured as a Domain Controller for the ezekiel.lab domain. Server Manager confirms that Active Directory Domain Services, DHCP Server, and DNS Server are all installed and running. DC01 is ready to accept client computers joining the ezekiel.lab domain.

Configure DHCP Scope

Note:

A DHCP scope is a range of IP addresses that DC01 will automatically assign to computers that join the ezekiel.lab domain. I created a scope in the 192.168.10.0 network range. Client computers will receive IP addresses between 192.168.10.100 and 192.168.10.200, leaving addresses below 100 reserved for servers and network devices with static IPs.



New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

< Back Next > Cancel

Note: I created a new DHCP scope named `ezekiel.lab Scope`. A scope name helps identify the purpose of the IP address range. The description explains that this scope is for client computers joining the `ezekiel.lab` domain.

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address:

End IP address:

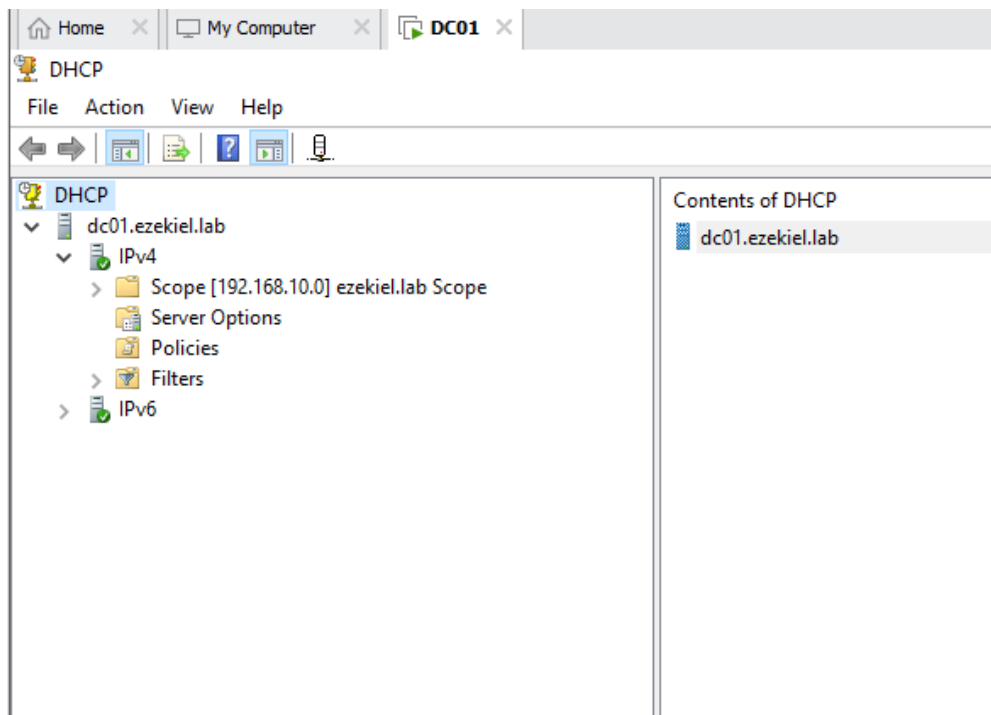
Configuration settings that propagate to DHCP Client

Length:

Subnet mask:

< Back Next > Cancel

Note: I configured the DHCP scope IP range from `192.168.10.100` to `192.168.10.200`. This gives DHCP 101 IP addresses to assign to client computers. Addresses below 100 are reserved for servers and network devices with static IPs like `DC01` at `192.168.10.10`.



DHCP Scope Created

Note:

The DHCP scope was created successfully. The scope named ezekiel.lab Scope covers the IP range 192.168.10.100 to 192.168.10.200. Any client computer that joins the ezekiel.lab domain will automatically receive an IP address from this range along with the correct gateway and DNS settings.